Unit 4 Summary: Scatterplots KEY

Name _____

Describing a Scatterplot

A **scatterplot** shows two numbers for each person or thing. The first number (x) goes across. The second number (y) goes up.

Feature	Ask	Example: the study data
Direction	As x goes up, does y go up (**positive**) or down (**negative**)? No pattern at all is **no association**.	**Positive.** The students who studied more mostly scored higher.
Form	Do the dots follow a line (**linear**) or a curve?	**Linear.** The dots follow a line.
Strength	Are the dots close to a line (**strong**), a little spread out (**moderate**), or very spread out (**weak**)?	**Strong.** The dots are close to the line.
Unusual	Any outliers or clusters?	None.

The Line of Best Fit

A **line of best fit** follows the trend of the dots. It goes through the middle of the dots. About the same number of dots are above it as below it.

Step	How it works	Example: the study data
1. Pick two points	Pick two points **on the line**. They do not have to be dots.	$(2, 64)$ and $(6, 84)$
2. Find the slope m	Subtract the y values. Subtract the x values in the same order. Divide.	$m = \frac{84 - 64}{6 - 2} = \frac{20}{4} = 5$
3. Find the y-intercept b	The equation is $\hat{y} = mx + b$. Put in m and one point. Solve for b.	$64 = 5(2) + b$, so $b = 54$
4. Write the equation	Put m and b into $\hat{y} = mx + b$. $\hat{y}$ is read "y-hat." The hat means **predicted**.	$\hat{y} = 5x + 54$
5. Predict	Put the x value into the equation. The line predicts y. It does not give the exact y.	A student studies 5 hours. $\hat{y} = 5(5) + 54 = 79$. The predicted score is **79**.
Extrapolation	Predicting far outside the data. It is risky.	12 hours: $\hat{y} = 5(12) + 54 = 114$. A test score cannot be more than 100. The data only goes from 1 to 8 hours.

Slope and Intercept in Context

	Sentence frame	**A plant:** $\hat{y} = 2.5x + 4$	**A candle:** $\hat{y} = -2x + 24$
Slope	For each additional [x unit], the predicted [y] increases (or decreases) by [slope and units].	For each additional **week**, the predicted **height** increases by **2.5 cm**.	For each additional **hour**, the predicted **height** decreases by **2 cm**.
y-intercept	When [x] is 0, the predicted [y] is [intercept and units].	When $x = 0$ (the day it was planted), the predicted height is **4 cm**.	When $x = 0$ (a new candle), the predicted height is **24 cm**.

Some intercepts have no meaning: if x is a person's height, then $x = 0$ is not possible.

Correlation and Causation

Idea	What it means	Example
Correlation coefficient r	It measures how strong a **linear** pattern is. r is always from -1 to 1.	The candles: $r \approx -0.98$. The dots are close to a line that goes down.
Sign of r	The sign shows the direction. $r > 0$ is positive. $r < 0$ is negative.	$r \approx 0.95$ is positive. $r \approx -0.80$ is negative.
Size of r	Close to 1 or -1 is strong. Close to 0 is weak.	$r = -0.8$ is stronger than $r = 0.6$. It is closer to -1 than 0.6 is to 1. The minus sign only tells the direction.
Correlation is not causation	A correlation shows that two things change together. It does **not** prove that one causes the other.	Ice cream sales and sunburns have a strong positive correlation. Ice cream does not cause sunburns.
Lurking variable	A hidden cause of both things.	Hot, sunny weather causes **both**.

Residuals

Idea	What it means	Example: the study line $\hat{y} = 5x + 54$
Residual	residual = actual $-$ predicted = $y - \hat{y}$	A student studied 4 hours and scored 78. $\hat{y} = 5(4) + 54 = 74$. Residual $= 78 - 74 = 4$.
Positive residual	The point is **above** the line.	Residual 4: the student scored 4 points **more** than predicted.
Negative residual	The point is **below** the line.	Another student studied 4 hours and scored 72. Residual $= 72 - 74 = -2$. The student scored 2 points **less** than predicted.

Least Squares and Residual Plots

Tool	How it works	What it means
Least-squares regression line	Square every residual and add them up: $\Sigma(y - \hat{y})^2$. Σ means add up.	This line makes the sum as **small as possible**. Calculators and computers find this line for you.
Residual plot	Put x across the bottom and the residuals up the side.	Random scatter, no pattern: a line is a **good** model. A curved pattern: a line is **not** a good model. The data is curved.